

ANKIT KUMAR

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PROFESSIONAL SUMMARY

MCA graduate and Full-Stack Developer with Generative AI expertise, skilled in building end-to-end web applications using React.js, Node.js, Express.js, and MongoDB, and integrating LLM-powered features including retrieval-augmented generation (RAG), vector search, and agentic tool routing using LangChain and Ollama. Experienced in independently designing, building, and deploying complete projects from database schema to production-ready UI, including real-time WebSocket-driven systems.

TECHNICAL SKILLS

Languages: JavaScript, Python

Frontend: React.js, Vite, HTML5, CSS3

Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets

Databases: MongoDB, Qdrant (Vector Database)

Generative AI / LLM: RAG (Retrieval-Augmented Generation), LangChain, Ollama, Prompt Engineering, Embeddings (Qwen3-Embedding), Agentic Tool Routing

AI/ML: TensorFlow/Keras, LSTM Neural Networks, MediaPipe, Computer Vision

Tools & Platforms: Git, GitHub, bcrypt, Express-Session

Version Control: Git, GitHub

PROJECTS

Stacks — Full Stack RAG Chatbot / *React, Vite, Express.js, Node.js, MongoDB, Qdrant, Ollama*

[GitHub](#)

- Engineered a full-stack retrieval-augmented generation (RAG) chatbot end-to-end solo, using a React/Vite frontend and an Express/Node.js backend, with Ollama-hosted Gemma for response generation and Qwen3-Embedding for semantic search.
- Implemented vector-based document retrieval with Qdrant, enabling context-aware, grounded chatbot responses over stored embeddings.
- Built secure user authentication using bcrypt password hashing, express-session, and connect-mongo for persistent session storage in MongoDB.
- Designed lazy conversation persistence and a sidebar interface supporting conversation rename/delete and theme persistence, improving usability across sessions.
- <https://stack-rag-chat-bot.vercel.app/>

Real Time Sign Language to Speech System / *Python, MediaPipe, TensorFlow/Keras (LSTM), LangChain, Ollama, PyQt5*

[GitHub](#)

- Built a real-time, full-stack computer vision system that classifies 21 hand gesture classes from a custom-collected dataset, using a stacked LSTM model trained on MediaPipe hand landmarks.
- Architected a FastAPI backend with WebSocket streaming (/ws/infer, /ws/collect) to run live inference and browser-driven data collection on frames sent from a React frontend, replacing the original desktop client entirely.
- Integrated LangChain with a locally hosted Ollama model to convert classified gesture sequences into coherent natural-language sentences in real time.
- Implemented server-side text-to-speech synthesis streamed back to the browser as audio, enabling live gesture-to-speech conversion end-to-end over the web.
- Achieved 100% classification accuracy on a custom 21-class test set through iterative data collection and model tuning.
- <https://real-time-sign-language-to-speech-s-tan.vercel.app/>

AI Voice Assistant / *Python, LangChain, Ollama, Tool Dispatch Routing*

[GitHub](#)

- Architected an agentic AI voice assistant with a tool-dispatch and routing layer that dynamically selects and invokes the correct function based on natural-language voice commands.
- Integrated LangChain with a locally hosted Ollama model to interpret user intent and route requests to the appropriate tool or action.
- Implemented a full voice input/output pipeline, enabling hands-free, on-device interaction with the assistant.

EDUCATION

Master of Computer Applications (MCA) — Graphic Era University, Dehradun

2024 – 2026

Bachelor of Computer Applications (BCA) — Birla Institute of Technology

2021 – 2024

CERTIFICATIONS

The Complete Full-Stack Web Development Bootcamp — Udemy (Dr. Angela Yu), 62 hours — Aug 2026

Claude Code in Action — Anthropic Education — May 2026